

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Lambert W Gaussian Integral

Evaluate the definite integral:

$$\int_0^{\infty} \frac{W(x)}{x} e^{-(W(x))^2} dx$$

where $W(x)$ is the Lambert W function satisfying $W(x)e^{W(x)} = x$.

Solution: Let the given integral be denoted as I :

$$I = \int_0^{\infty} \frac{W(x)}{x} e^{-(W(x))^2} dx$$

We perform a substitution using the defining relation of the Lambert W function. Let $u = W(x)$. By definition, we have:

$$x = ue^u$$

Differentiating both sides with respect to u , we apply the product rule to find the differential dx :

$$dx = (e^u + ue^u) du = (1 + u)e^u du$$

Next, we determine the new limits of integration:

- As $x \rightarrow 0^+$, since $0 = 0 \cdot e^0$, we have $u = W(0) \rightarrow 0$.
- As $x \rightarrow \infty$, since ue^u grows unbounded, we have $u = W(x) \rightarrow \infty$.

Substituting $W(x) = u$, $x = ue^u$, and $dx = (1 + u)e^u du$ into the integral, we obtain:

$$I = \int_0^{\infty} \frac{u}{ue^u} e^{-u^2} (1 + u)e^u du$$

Notice how the term ue^u in the denominator completely cancels with the prefactor ue^u in the numerator and differential:

$$I = \int_0^{\infty} (1 + u)e^{-u^2} du$$

We split this integral into the sum of two standard integrals:

$$I = \int_0^{\infty} e^{-u^2} du + \int_0^{\infty} ue^{-u^2} du$$

The first term is the familiar Gaussian integral over the positive half-line:

$$\int_0^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2}$$

The second term can be evaluated directly via the substitution $v = u^2 \implies dv = 2u du$:

$$\int_0^\infty ue^{-u^2} du = \left[-\frac{1}{2}e^{-u^2} \right]_0^\infty = 0 - \left(-\frac{1}{2} \right) = \frac{1}{2}$$

Combining both evaluations yields the final exact value:

$$I = \frac{\sqrt{\pi} + 1}{2}$$

1.2 Hard Integral

Nested Logarithmic Integral

Evaluate the definite integral:

$$\int_0^\infty \frac{\ln(e^x + \ln(e^x + \dots))}{e^x + \ln(e^x + \ln(e^x + \dots))} dx$$

Solution: Let the infinite nested logarithmic expression be denoted by the function $f(x)$:

$$f(x) = \ln(e^x + \ln(e^x + \dots))$$

By observing the self-similar structure of the nested expression, we can write the functional equation:

$$f(x) = \ln(e^x + f(x))$$

Exponentiating both sides gives an algebraic relation between $f(x)$ and x :

$$e^{f(x)} = e^x + f(x) \implies e^{f(x)} - f(x) = e^x$$

Notice that the integrand in our original problem can be written directly in terms of $f(x)$:

- The numerator is simply $f(x)$.
- The denominator is $e^x + f(x)$, which equals $e^{f(x)}$.

Therefore, the integral simplifies to:

$$I = \int_0^\infty \frac{f(x)}{e^{f(x)}} dx$$

We now perform the substitution $u = f(x)$. To express dx in terms of u , we differentiate the relation $e^u - u = e^x$ with respect to x :

$$(e^u - 1) \frac{du}{dx} = e^x \implies dx = \frac{e^u - 1}{e^x} du = \frac{e^u - 1}{e^u - u} du$$

Let us determine the boundaries for the new variable u :

- At $x = 0$, the relation becomes $e^u - u = e^0 = 1$. The unique real solution is $u = 0$.

- As $x \rightarrow \infty$, since $e^x \rightarrow \infty$, we must have $u \rightarrow \infty$.

Transforming the integral into the u -domain gives:

$$I = \int_0^\infty \frac{u}{e^u} \left(\frac{e^u - 1}{e^u - u} \right) du = \int_0^\infty \frac{u(1 - e^{-u})}{e^u - u} du$$

To evaluate this integral, we expand the denominator into a geometric series since $ue^{-u} < 1$ for all $u > 0$:

$$\frac{1}{e^u - u} = \frac{e^{-u}}{1 - ue^{-u}} = \sum_{n=1}^{\infty} u^{n-1} e^{-nu}$$

Substituting this series expansion back into the integrand:

$$\begin{aligned} I &= \int_0^\infty u(1 - e^{-u}) \sum_{n=1}^{\infty} u^{n-1} e^{-nu} du = \int_0^\infty (1 - e^{-u}) \sum_{n=1}^{\infty} u^n e^{-nu} du \\ I &= \sum_{n=1}^{\infty} \int_0^\infty u^n e^{-nu} du - \sum_{n=1}^{\infty} \int_0^\infty u^n e^{-(n+1)u} du \end{aligned}$$

Using the standard Gamma function integral $\int_0^\infty t^k e^{-at} dt = \frac{k!}{a^{k+1}}$, we evaluate both summations term-by-term:

$$I = \sum_{n=1}^{\infty} \frac{n!}{n^{n+1}} - \sum_{n=1}^{\infty} \frac{n!}{(n+1)^{n+1}}$$

We write out the first few terms of this series explicitly to observe its behavior:

$$I = \left(\frac{1!}{1^2} + \frac{2!}{2^3} + \frac{3!}{3^4} + \dots \right) - \left(\frac{1!}{2^2} + \frac{2!}{3^3} + \frac{3!}{4^4} + \dots \right)$$

Notice that for the second summation, the term corresponding to index n is $\frac{n!}{(n+1)^{n+1}}$. In the first summation, the term corresponding to index $n+1$ is:

$$\frac{(n+1)!}{(n+1)^{n+2}} = \frac{(n+1) \cdot n!}{(n+1)^{n+2}} = \frac{n!}{(n+1)^{n+1}}$$

Thus, every term in the second series perfectly cancels with the subsequent term of the first series! This is a telescoping series where all intermediate terms vanish, leaving only the very first term of the first sum:

$$I = \frac{1!}{1^2} - \lim_{N \rightarrow \infty} \frac{N!}{(N+1)^{N+1}} = 1 - 0 = 1$$

The exact value of the integral is:

$$I = 1$$

Differentials

2.1 Medium Differential

Second-Order Variable Coefficient Equation

Consider the second-order differential equation:

$$xy'' + 2y' = 12x^2$$

Given the initial condition $f(0) = 1$, compute the value of $f(2)$. (Assume the function $f(x)$ and its derivatives are continuous.)

Solution: We begin by multiplying both sides of the differential equation by the independent variable x :

$$x^2y'' + 2xy' = 12x^3$$

We recognize that the left-hand side is the exact derivative of the product x^2y' with respect to x :

$$\frac{d}{dx} (x^2y') = x^2y'' + 2xy'$$

Thus, the differential equation can be rewritten in exact form:

$$\frac{d}{dx} (x^2y') = 12x^3$$

Integrating both sides with respect to x yields:

$$x^2y' = \int 12x^3 dx = 3x^4 + C_1$$

Dividing through by x^2 gives the first derivative:

$$y'(x) = 3x^2 + \frac{C_1}{x^2}$$

We are given that the function $f(x)$ and its derivatives are continuous everywhere, including at $x = 0$. For $y'(0)$ to be finite and well-defined, the term $\frac{C_1}{x^2}$ must not exhibit a singularity at the origin. Therefore, we must have:

$$C_1 = 0$$

This simplifies our first derivative to:

$$y'(x) = 3x^2$$

Integrating a second time with respect to x , we find the general solution:

$$y(x) = \int 3x^2 dx = x^3 + C_2$$

Using the initial condition $f(0) = 1$, we determine the constant of integration C_2 :

$$y(0) = 0^3 + C_2 = 1 \implies C_2 = 1$$

Therefore, the unique continuous solution to the differential equation is:

$$f(x) = x^3 + 1$$

Finally, evaluating this function at $x = 2$:

$$f(2) = 2^3 + 1 = 8 + 1 = 9$$

2.2 Hard Differential

System of Non-Homogeneous Differential Equations

Solve the system of non-homogeneous linear differential equations:

$$\begin{cases} x'(t) = -4x(t) - y(t) + e^{-t} \\ y'(t) = x(t) - 2y(t) + 2e^{-3t} \end{cases}$$

Given the initial conditions $x(0) = 2$ and $y(0) = -2$, find $x(1)$.

Solution: We solve this coupled system using the method of systematic elimination. From the second differential equation, we isolate $x(t)$:

$$x(t) = y'(t) + 2y(t) - 2e^{-3t}$$

Differentiating this expression with respect to t gives $x'(t)$:

$$x'(t) = y''(t) + 2y'(t) + 6e^{-3t}$$

We now substitute both $x(t)$ and $x'(t)$ into the first differential equation:

$$y''(t) + 2y'(t) + 6e^{-3t} = -4(y'(t) + 2y(t) - 2e^{-3t}) - y(t) + e^{-t}$$

Expanding and collecting all terms involving $y(t)$ and its derivatives on the left-hand side:

$$\begin{aligned} y''(t) + 2y'(t) + 6e^{-3t} &= -4y'(t) - 8y(t) + 8e^{-3t} - y(t) + e^{-t} \\ y''(t) + 6y'(t) + 9y(t) &= 2e^{-3t} + e^{-t} \end{aligned}$$

This is a second-order linear non-homogeneous differential equation with constant coefficients. We first solve the associated homogeneous equation:

$$r^2 + 6r + 9 = 0 \implies (r + 3)^2 = 0$$

Since $r = -3$ is a repeated root, the complementary homogeneous solution is:

$$y_h(t) = (C_1 + C_2 t)e^{-3t}$$

To find the particular solution $y_p(t)$, we use the method of undetermined coefficients. Since -3 is a double root of the characteristic equation, for the term $2e^{-3t}$ we must multiply our trial solution by t^2 . For the term e^{-t} , a standard exponential trial function suffices:

$$y_p(t) = At^2e^{-3t} + Be^{-t}$$

Differentiating the trial solution:

$$y'_p(t) = A(2t - 3t^2)e^{-3t} - Be^{-t}$$

$$y''_p(t) = A(2 - 12t + 9t^2)e^{-3t} + Be^{-t}$$

Substituting these into the left-hand side of our second-order equation:

$$\begin{aligned} y''_p + 6y'_p + 9y_p &= Ae^{-3t}[(2 - 12t + 9t^2) + 6(2t - 3t^2) + 9t^2] + Be^{-t}[1 - 6 + 9] \\ &= 2Ae^{-3t} + 4Be^{-t} \end{aligned}$$

Equating coefficients with the right-hand side $2e^{-3t} + e^{-t}$:

$$2A = 2 \implies A = 1, \quad 4B = 1 \implies B = \frac{1}{4}$$

Thus, the general solution for $y(t)$ is:

$$y(t) = (C_1 + C_2t + t^2)e^{-3t} + \frac{1}{4}e^{-t}$$

Using our algebraic relation for $x(t)$, we compute $y'(t) + 2y(t)$:

$$y'(t) = [(C_2 - 3C_1) + (2 - 3C_2)t - 3t^2]e^{-3t} - \frac{1}{4}e^{-t}$$

$$2y(t) = [2C_1 + 2C_2t + 2t^2]e^{-3t} + \frac{1}{2}e^{-t}$$

Adding these together and subtracting $2e^{-3t}$:

$$x(t) = y'(t) + 2y(t) - 2e^{-3t} = [(C_2 - C_1 - 2) + (2 - C_2)t - t^2]e^{-3t} + \frac{1}{4}e^{-t}$$

We now apply the initial conditions at $t = 0$:

$$y(0) = C_1 + \frac{1}{4} = -2 \implies C_1 = -\frac{9}{4}$$

$$x(0) = C_2 - C_1 - 2 + \frac{1}{4} = 2 \implies C_2 - \left(-\frac{9}{4}\right) - \frac{7}{4} = 2 \implies C_2 + \frac{1}{2} = 2 \implies C_2 = \frac{3}{2}$$

Substituting $C_1 = -\frac{9}{4}$ and $C_2 = \frac{3}{2}$ into the coefficients of $x(t)$:

$$C_2 - C_1 - 2 = \frac{3}{2} + \frac{9}{4} - 2 = \frac{7}{4}, \quad 2 - C_2 = 2 - \frac{3}{2} = \frac{1}{2}$$

Therefore, the exact time-dependent solution for $x(t)$ is:

$$x(t) = \left(\frac{7}{4} + \frac{1}{2}t - t^2\right)e^{-3t} + \frac{1}{4}e^{-t}$$

Evaluating at $t = 1$:

$$x(1) = \left(\frac{7}{4} + \frac{1}{2} - 1\right)e^{-3} + \frac{1}{4}e^{-1} = \frac{5}{4}e^{-3} + \frac{1}{4}e^{-1} = \frac{5e^{-3} + e^{-1}}{4}$$

Derivatives

3.1 Medium Derivative

Trigonometric Inverse Exponent Derivative

Let

$$f(x) = \left[\tan^{-1} \left(\frac{(\sin(x))^{\ln(e^2)} + \cos^2(x)}{\sqrt{\csc^2(x) - 1}} \right) \right]^{\ln(x)}$$

Find $f'(1) + f''(1)$.

Solution: We begin by drastically simplifying the argument of the function $f(x)$ using standard trigonometric identities. First, examine the exponent in the numerator:

$$\ln(e^2) = 2 \ln(e) = 2$$

Thus, the numerator inside the inverse tangent function simplifies via the Pythagorean identity:

$$(\sin(x))^{\ln(e^2)} + \cos^2(x) = \sin^2(x) + \cos^2(x) = 1$$

Next, examine the denominator inside the inverse tangent function:

$$\csc^2(x) - 1 = \cot^2(x) \implies \sqrt{\csc^2(x) - 1} = \sqrt{\cot^2(x)} = \cot(x)$$

(Note: since we are evaluating derivatives at $x = 1$ radian $\approx 57.3^\circ$, which lies in the first quadrant $(0, \pi/2)$, $\cot(x)$ is strictly positive). Substituting these simplified terms back into the inverse tangent function:

$$\tan^{-1} \left(\frac{1}{\cot(x)} \right) = \tan^{-1}(\tan(x)) = x$$

Therefore, the entire complicated expression defining $f(x)$ collapses to a remarkably simple power-exponential function:

$$f(x) = x^{\ln(x)} = e^{(\ln(x))^2}$$

We now compute the first derivative $f'(x)$ using the chain rule:

$$f'(x) = \frac{d}{dx} e^{(\ln(x))^2} = e^{(\ln(x))^2} \cdot \frac{d}{dx} (\ln^2(x)) = x^{\ln(x)} \cdot \left(\frac{2 \ln(x)}{x} \right) = 2x^{\ln(x)-1} \ln(x)$$

Evaluating the function and its first derivative at $x = 1$:

$$f(1) = 1^{\ln(1)} = 1^0 = 1$$

$$f'(1) = 2(1)^{0-1} \ln(1) = 2(1)(0) = 0$$

Next, we differentiate $f'(x) = 2 \ln(x) \cdot x^{\ln(x)-1}$ using the product rule to find the second derivative $f''(x)$:

$$f''(x) = 2 \left[\frac{d}{dx} (\ln(x)) \cdot x^{\ln(x)-1} + \ln(x) \cdot \frac{d}{dx} (x^{\ln(x)-1}) \right]$$

$$f''(x) = 2 \left[\frac{1}{x} \cdot x^{\ln(x)-1} + \ln(x) \cdot \frac{d}{dx} \left(x^{\ln(x)-1} \right) \right]$$

When evaluating $f''(1)$, notice that since $\ln(1) = 0$, the second term inside the bracket vanishes completely regardless of the derivative of $x^{\ln(x)-1}$:

$$f''(1) = 2 \left[\frac{1}{1} \cdot 1^{0-1} + 0 \right] = 2(1) = 2$$

Summing the two derivatives yields the final result:

$$f'(1) + f''(1) = 0 + 2 = 2$$

3.2 Hard Derivative

Infinite Arctangent Series Derivative

Let

$$f(x) = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{x}{n^2} \right)$$

Find $f'(1/2)$.

Solution: To find the derivative of the infinite series, we differentiate term-by-term with respect to x :

$$f'(x) = \sum_{n=1}^{\infty} \frac{d}{dx} \tan^{-1} \left(\frac{x}{n^2} \right) = \sum_{n=1}^{\infty} \frac{1}{1 + \left(\frac{x}{n^2} \right)^2} \cdot \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{n^2}{n^4 + x^2}$$

We are required to evaluate this derivative at $x = 1/2$:

$$f'(1/2) = \sum_{n=1}^{\infty} \frac{n^2}{n^4 + 1/4} = \sum_{n=1}^{\infty} \frac{4n^2}{4n^4 + 1}$$

To evaluate this series algebraically, we factor the denominator using Sophie Germain's identity, $4n^4 + 1 = (2n^2 - 2n + 1)(2n^2 + 2n + 1)$, and perform partial fraction decomposition:

$$\frac{4n^2}{4n^4 + 1} = \frac{n}{2n^2 - 2n + 1} - \frac{n}{2n^2 + 2n + 1}$$

By completing the square in each denominator, we can express the terms as:

$$\frac{n}{2n^2 - 2n + 1} = \frac{2n}{(2n - 1)^2 + 1}, \quad \frac{n}{2n^2 + 2n + 1} = \frac{2n}{(2n + 1)^2 + 1}$$

Substituting this back into the summation yields:

$$f'(1/2) = \sum_{n=1}^{\infty} \left[\frac{2n}{(2n - 1)^2 + 1} - \frac{2n}{(2n + 1)^2 + 1} \right]$$

We write out the first few terms of this series to observe its structural behavior:

- For $n = 1$: $\frac{2(1)}{1^2+1} - \frac{2(1)}{3^2+1} = 1 - \frac{2}{3^2+1}$
- For $n = 2$: $\frac{2(2)}{3^2+1} - \frac{2(2)}{5^2+1} = \frac{4}{3^2+1} - \frac{4}{5^2+1}$
- For $n = 3$: $\frac{2(3)}{5^2+1} - \frac{2(3)}{7^2+1} = \frac{6}{5^2+1} - \frac{6}{7^2+1}$

Notice how consecutive terms combine across shared denominators of the form $(2k + 1)^2 + 1$:

$$-\frac{2k}{(2k+1)^2+1} + \frac{2(k+1)}{(2k+1)^2+1} = \frac{2}{(2k+1)^2+1}$$

This pairwise cancellation collapses the entire infinite series into a sum over all odd positive integers:

$$f'(1/2) = 1 + \sum_{k=1}^{\infty} \frac{2}{(2k+1)^2+1} = \sum_{k=0}^{\infty} \frac{2}{(2k+1)^2+1} = \sum_{m \text{ odd} > 0} \frac{2}{m^2+1}$$

We evaluate this odd-integer series by taking the difference between the sum over all positive integers and the sum over all even positive integers ($m = 2j$), utilizing the standard hyperbolic Mittag-Leffler series $\sum_{n=1}^{\infty} \frac{1}{n^2+a^2} = \frac{\pi a \coth(\pi a) - 1}{2a^2}$:

$$\sum_{n=1}^{\infty} \frac{2}{n^2+1} = \pi \coth(\pi) - 1$$

$$\sum_{j=1}^{\infty} \frac{2}{(2j)^2+1} = \frac{1}{2} \sum_{j=1}^{\infty} \frac{1}{j^2 + (1/2)^2} = \frac{\pi}{2} \coth\left(\frac{\pi}{2}\right) - 1$$

Subtracting the even terms from the total sum leaves:

$$f'(1/2) = [\pi \coth(\pi) - 1] - \left[\frac{\pi}{2} \coth\left(\frac{\pi}{2}\right) - 1 \right] = \pi \coth(\pi) - \frac{\pi}{2} \coth\left(\frac{\pi}{2}\right)$$

Using the double-angle hyperbolic cotangent identity $\coth(\pi) = \frac{1}{2} \coth\left(\frac{\pi}{2}\right) + \frac{1}{2} \tanh\left(\frac{\pi}{2}\right)$, this expression simplifies to its final exact closed form:

$$f'(1/2) = \frac{\pi}{2} \tanh\left(\frac{\pi}{2}\right) \approx 1.440636$$

Physics & Engineering

4.1 Mechanics

Orca Wedge Lift Dynamics

A 4,000 kg orca is resting in a symmetric V-shaped steel cradle. The walls of the cradle form an angle of 60° with each other (meaning 30° on each side of the vertical centerline). The coefficient of static friction between the orca and the wet steel is $\mu_s = 0.05$. Calculate the vertical upward force F_{lift} (in N) required to pull the orca out of the wedge. Take $g = 9.81 \text{ m s}^{-2}$.

Solution: Let $\theta = 30^\circ$ be the inclination angle of each cradle wall with respect to the vertical centerline. We first analyze the mechanics of self-locking wedges. When an upward vertical lifting force F_{lift} is applied to an object inside a rigid symmetric wedge, the impending motion of the object is directed vertically upwards. Consequently, static friction forces f_s act downward along the inclined walls to oppose this motion.

Let N be the magnitude of the normal contact force exerted by each wall on the orca. By symmetry, horizontal forces automatically balance ($N \cos \theta = N \cos \theta$). The equation for vertical equilibrium at the threshold of upward motion is:

$$F_{\text{lift}} + 2N \sin \theta = W + 2f_s \cos \theta$$

where $W = mg$ is the gravitational weight of the orca. Substituting the maximum static friction condition $f_s = \mu_s N$:

$$F_{\text{lift}} + 2N \sin \theta = W + 2\mu_s N \cos \theta \implies F_{\text{lift}} = W + 2N (\mu_s \cos \theta - \sin \theta)$$

Notice the critical term $(\mu_s \cos \theta - \sin \theta)$. If $\mu_s \geq \tan \theta$, the wedge exhibits ****self-locking**** (or mechanical jamming), meaning that pulling upward requires additional force to overcome frictional binding against the walls ($N > 0$).

We evaluate the self-locking threshold for our specific geometry:

$$\tan(30^\circ) = \frac{1}{\sqrt{3}} \approx 0.577$$

Since the given friction coefficient $\mu_s = 0.05$ is strictly less than $\tan(30^\circ) \approx 0.577$, the V-shaped cradle is ****non-self-locking****.

Physically, this means that as the upward lifting force F_{lift} increases from zero, it directly relieves the gravitational load pressing the orca against the walls. The normal contact forces N decrease continuously and smoothly to zero without any frictional wedging or binding. At the exact threshold where lifting occurs ($N \rightarrow 0$), the required upward force is simply equal to the total weight of the orca:

$$F_{\text{lift}} = mg$$

Substituting the numerical values provided:

$$F_{\text{lift}} = 4000 \text{ kg} \times 9.81 \text{ m s}^{-2} = 39,240 \text{ N}$$

4.2 Quantum Mechanics

Harmonic Oscillator Quartic Position Expectation

A quantum harmonic oscillator is in its first excited state $|1\rangle$. The lowering and raising operators $\hat{a}$ and $\hat{a}^\dagger$ satisfy $[\hat{a}, \hat{a}^\dagger] = 1$. The position operator $\hat{x}$ is defined as:

$$\hat{x} = \frac{1}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger)$$

Evaluate the exact numerical value of the expectation value:

$$I = \langle 1 | \hat{x}^4 | 1 \rangle$$

Assume the states are normalized such that $\langle n | n \rangle = 1$.

Solution: Using the definition of the position operator, we express $\hat{x}^4$ in terms of the ladder operators:

$$\hat{x}^4 = \frac{1}{4} (\hat{a} + \hat{a}^\dagger)^4$$

Thus, the expectation value becomes:

$$I = \frac{1}{4} \langle 1 | (\hat{a} + \hat{a}^\dagger)^4 | 1 \rangle$$

Since the operator $(\hat{a} + \hat{a}^\dagger)^2$ is Hermitian, we can rewrite the inner matrix element as the squared norm of an intermediate state:

$$\langle 1 | (\hat{a} + \hat{a}^\dagger)^4 | 1 \rangle = \left\| (\hat{a} + \hat{a}^\dagger)^2 | 1 \rangle \right\|^2$$

We compute the action of $(\hat{a} + \hat{a}^\dagger)$ on $|1\rangle$ sequentially using the fundamental ladder operator relations $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$:

Step 1: First application:

$$(\hat{a} + \hat{a}^\dagger) |1\rangle = \hat{a}|1\rangle + \hat{a}^\dagger|1\rangle = \sqrt{1}|0\rangle + \sqrt{2}|2\rangle = |0\rangle + \sqrt{2}|2\rangle$$

Step 2: Second application:

$$(\hat{a} + \hat{a}^\dagger)^2 |1\rangle = (\hat{a} + \hat{a}^\dagger) (|0\rangle + \sqrt{2}|2\rangle)$$

We distribute the operators across the superposition:

$$(\hat{a} + \hat{a}^\dagger) |0\rangle = \hat{a}|0\rangle + \hat{a}^\dagger|0\rangle = 0 + |1\rangle = |1\rangle$$

$$(\hat{a} + \hat{a}^\dagger) (\sqrt{2}|2\rangle) = \sqrt{2} (\hat{a}|2\rangle + \hat{a}^\dagger|2\rangle) = \sqrt{2} (\sqrt{2}|1\rangle + \sqrt{3}|3\rangle) = 2|1\rangle + \sqrt{6}|3\rangle$$

Combining these results yields the resulting normalized state vector:

$$|\psi_2\rangle = (\hat{a} + \hat{a}^\dagger)^2 |1\rangle = |1\rangle + 2|1\rangle + \sqrt{6}|3\rangle = 3|1\rangle + \sqrt{6}|3\rangle$$

Step 3: Compute the squared norm: Since the Fock states $|1\rangle$ and $|3\rangle$ are orthonormal ($\langle 1|3\rangle = 0$):

$$\langle \psi_2 | \psi_2 \rangle = 3^2 + (\sqrt{6})^2 = 9 + 6 = 15$$

Therefore, substituting this inner product back into our expression for I :

$$I = \frac{1}{4}(15) = \frac{15}{4} = 3.75$$

4.3 Electromagnetism

RC Series Filament Lamp Circuit

A metal filament lamp is connected in series with a capacitor of capacitance $C \mu\text{F}$ across a 200 V, 50 Hz supply. The power consumed by the lamp is 500 W while the voltage drop across it is 100 V. Assume no inductive load. Find the ratio C/ϕ where ϕ is the phase angle magnitude in degrees.

Solution: We analyze the AC circuit step-by-step using root-mean-square (RMS) phasors:

Step 1: Determine Circuit Current and Lamp Resistance A metal filament lamp behaves as a purely resistive load R . Given that the real power consumed by the lamp is $P = 500 \text{ W}$ and the RMS voltage drop across it is $V_R = 100 \text{ V}$, the series circuit current I is:

$$I = \frac{P}{V_R} = \frac{500 \text{ W}}{100 \text{ V}} = 5 \text{ A}$$

The resistance of the filament lamp is therefore:

$$R = \frac{V_R}{I} = \frac{100 \text{ V}}{5 \text{ A}} = 20 \Omega$$

Step 2: Calculate Capacitor Voltage and Reactance In a series RC circuit across a total RMS supply voltage $V_{\text{total}} = 200 \text{ V}$, the resistive voltage drop V_R and capacitive voltage drop V_C are in quadrature (90° out of phase). By the phasor voltage triangle:

$$V_{\text{total}}^2 = V_R^2 + V_C^2 \implies V_C = \sqrt{V_{\text{total}}^2 - V_R^2}$$

$$V_C = \sqrt{200^2 - 100^2} = \sqrt{40000 - 10000} = \sqrt{30000} = 100\sqrt{3} \text{ V}$$

Using Ohm's law for AC reactance, the capacitive reactance X_C is:

$$X_C = \frac{V_C}{I} = \frac{100\sqrt{3} \text{ V}}{5 \text{ A}} = 20\sqrt{3} \Omega$$

Step 3: Determine Phase Angle Magnitude ϕ The phase angle ϕ between the total supply voltage and the circuit current is related to the impedance ratio:

$$\tan \phi = \frac{X_C}{R} = \frac{20\sqrt{3} \Omega}{20 \Omega} = \sqrt{3}$$

Taking the inverse tangent yields the exact magnitude in degrees:

$$\varphi = 60^\circ$$

Step 4: Calculate Capacitance C The capacitive reactance is defined by $X_C = \frac{1}{2\pi fC}$, where supply frequency $f = 50$ Hz. Solving for capacitance C in Farads:

$$C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi(50)(20\sqrt{3})} = \frac{1}{2000\pi\sqrt{3}} \text{ F}$$

Converting to microfarads (μF) by multiplying by 10^6 :

$$C(\text{in } \mu\text{F}) = \frac{10^6}{2000\pi\sqrt{3}} = \frac{500}{\pi\sqrt{3}} \mu\text{F} \approx 91.888 \mu\text{F}$$

Step 5: Compute the Required Ratio C/ φ Dividing the numerical capacitance value in μF by the phase angle magnitude in degrees ($\varphi = 60$):

$$\frac{C}{\varphi} = \frac{\frac{500}{\pi\sqrt{3}}}{60} = \frac{500}{60\pi\sqrt{3}} = \frac{25}{3\pi\sqrt{3}} \approx 1.531$$